

Odometry motion model:-

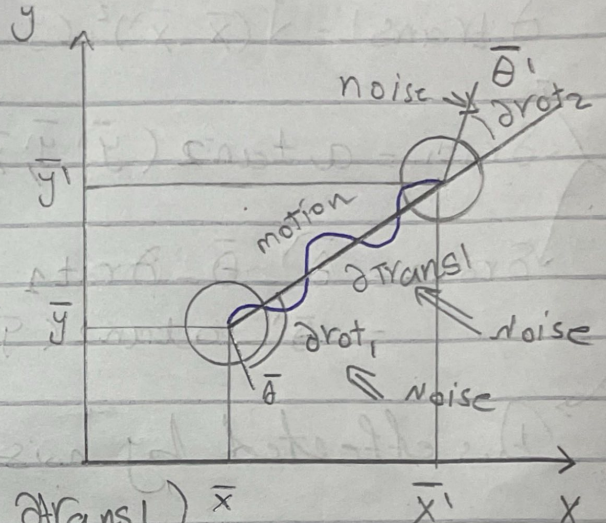
"Odometry tells us how the robot moved, but odometry is noisy. So instead of saying 'the robot moved exactly here', we generate many possible motions around the measured motion."

$$\Delta_{trans} = \sqrt{(\bar{x}' - \bar{x})^2 + (\bar{y}' - \bar{y})^2}$$

$$\Delta_{rot_1} = \arctan_2(\bar{y}' - \bar{y}, \bar{x}' - \bar{x}) - \bar{\theta}$$

$$\Delta_{rot_2} = \bar{\theta}' - \bar{\theta} - \Delta_{rot_1}$$

but each function has noise
we need to filterization.



$$\hat{\Delta}_{rot_1} = \Delta_{rot_1} - P(\alpha_1 \Delta_{rot_1} + \alpha_2 \Delta_{transl}) \bar{x}$$

$$\hat{\Delta}_{transl} = \Delta_{transl} - P(\alpha_3 \Delta_{transl} + \alpha_4 (\Delta_{rot_1} + \Delta_{rot_2}))$$

$$\hat{\Delta}_{rot_2} = \Delta_{rot_2} - P(\alpha_5 \Delta_{rot_2} + \alpha_6 \Delta_{transl})$$

Noise

Sol:-

$$\bar{x}' - \bar{x} = \hat{\Delta}_{transl} \cos(\bar{\theta})$$

$$\bar{y}' - \bar{y} = \hat{\Delta}_{transl} \sin(\bar{\theta})$$

$$\bar{\theta}' - \bar{\theta} = \hat{\Delta}_{rot_1} + \hat{\Delta}_{rot_2}$$

$$\bar{\theta} = \bar{\theta} + \hat{\Delta}_{rot_1}$$

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$$\begin{bmatrix} \bar{x}' \\ \bar{y}' \\ \bar{\theta}' \end{bmatrix} = \begin{bmatrix} \bar{x} \\ \bar{y} \\ \bar{\theta} \end{bmatrix} + \begin{bmatrix} \hat{\Delta}_{transl} \cos(\bar{\theta} + \hat{\Delta}_{rot_1}) \\ \hat{\Delta}_{transl} \sin(\bar{\theta} + \hat{\Delta}_{rot_1}) \\ \hat{\Delta}_{rot_1} + \hat{\Delta}_{rot_2} \end{bmatrix}$$

New
Pos

old
Pos

